



# BI Justification

## Digital Wellness for Tomorrow

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## EXECUTIVE SUMMARY

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### Project Overview

The Digital Wellness for Tomorrow initiative is a strategic Business Intelligence project led by I.Q. Productionz. It was designed to mitigate the negative economic and psychological impacts of excessive screen time and strengthen the positive ones. By integrating data from school surveys, wellness applications and the CDC, it is imperative that we can identify behavioral risk patterns in children and employees. In order to boost workforce productivity by an estimated 15%, providing data-driven, actionable interventions is our drive. These predictive models foster high-value partnerships within the EdTech sector.

### Strategic Assumptions

The project is built on the assumption that screen-time data from external wellness apps and school surveys can be successfully standardized and that stakeholders (HR, parents, and educators) will remain actively engaged in the intervention phase. We further assume that our current cloud-native infrastructure can scale to handle real-time data feeds with high integrity.

### Risks and Issues

A primary concern is **Data Quality Risk**, specifically regarding the inconsistency of self-reported screen-time metrics. Additionally, we must navigate **Interpretation Risk**, ensuring that data correlations are not misconstrued as absolute causation. Currently, the project faces a challenge regarding the limited availability of longitudinal datasets directly linking early screen habits to long-term employment outcomes, which will be addressed through iterative predictive modeling.

### Key Project Metrics

- **Timeline:** 6 Months (January 5 – July 7, 2025).
- **Budget:** \$57,000 (Total Cost of Ownership).
- **ROI:** Strongly positive, driven by a reduction in mental-health-related absenteeism and the elimination of manual reporting through automated BI dashboards.

### Recommendation

Based on the high strategic alignment with future workforce readiness and the significant potential for revenue growth through wellness innovation, it is the **strong recommendation of the project team to proceed** with the immediate initiation of Phase 1. The cost of inaction—including potential brand backlash and declining productivity—far outweighs the projected \$57,000 investment.

## BUSINESS OPPORTUNITIES

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The key opportunities and problems that this project targets is concentrated on driving returns on investment through analyzing screen time impacts on the young. The mental health risks, reputation concerns for businesses and productivity. Gaps have been linked to excessive screen time among children and employees. Using data science, we can leverage our future employees by implementing recommend interventions to at risk groups identified by screen times patterns.

Significant opportunities via managed screen time leads to boosting employee productivity, developing digital wellness solutions and innovating edtech partnerships. Depression prevails in kids even with more than two hours of screen time a day by 25.9%. Not only would we be helping children in their current state, but their future depends on their phycological health in order to thrive in society—including employment. Addressing digital wellness can protect our children and address brand backlash over digital addiction along with fatigue in remote work.

Harnessing insights from screen time can drive revenue, enhance productivity in the workforce and improve brand reputation. Maximizing, partnerships, and swift responses to risks support growth. Considering these objectives, brand trust strengthens, productivity is expected to boost by 15% and additional streams of revenue form.

Highly impactful businesses target analytical gaps. The strategic initiatives we will need to align this project involve committed businesses that will stay involved and provide ongoing sponsorship. We strive to fill the voids in productivity and reform digital risks. Active role-players encompass, HR teams, parents and educators. Our deliverables offer growing revenue and brand trust enhancement. Our first step will be to find BI initiatives that may already exist and learn from them.

### Business Objectives

Mitigating risks and driving growth for our businesses is the aim to achieve with insights from screen time use. Educators, HR teams and parents will need to implement the interventions that we recommend after successful discovery of at-risk groups through screen time pattern analyzation. The innovation of Ed Tech partnerships, our digital wellness solutions, and the productivity boost we can offer will be due to the screen time management procedures compiled in our strategies. Essentially evaluating revenue will be our success measurement indicator. Employee productivity could be a direct link to revenue. We expect to see an increase in uptime and a decrease mental health related absences due to the screen time management tools and the wellness-focused products/services we provide.

## Business Initiatives

Workforce productivity optimization and digital wellness Innovation are the key aspects our project supports. The screen time analytics delivered will enable us to enhance digital wellness tools to support employee wellness programs implemented through EdTech partnerships. Our recommendations can be personalized with predictive model use and empower HR teams, educators, and parents with actionable screen time mitigation techniques. The capabilities around digital wellness decisions will be enhanced through our project and optimize work and learning environments. Some of the tools we will enact to drive brand reputation and revenue growth are predictive risk models, screen time dashboards and intervention playbooks.

## Business Processes

The business processes our project supports are edtech partnerships, digital product development, and employee wellness programs. The possible bottlenecking that could hinder targeted interventions intertwine the lack of granular data, missed partnerships and slow risk responses. Insights into screen time patterns may be limited. If screen time impacts are unable to be quantified, edtech collaborations could be blocked. Identifying the risk reductions that impact productivity and Mental Health could be delayed. The actionable analytics, wellness-focused revenue streams and proactive outreach this project provides addresses these concerns.

## PROJECT DESCRIPTION

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Screen time analytics harnessed will drive business to grow with innovative digital wellness solutions. Risks will be mitigated by identifying at risk groups to recommend interventions to. After usage patterns are analyzed, edtech partnerships can deliver actionable insights to boost workforce productivity. Revenue streams are increased and mental health improved with dashboards that provide predictive models offering safe screen time limitation suggestions. Safer screen time solutions also enhance brand representation mitigating further sales.

## Project Goals

The actionable screen time insights created by this project expand edtech partnerships and employee wellness programs. Paired with predictive analytic tools, delivering dashboards for HR teams and parents are expected to drive success through proactive interventions. By optimizing work/learning environments, data-driven decision-making comes first-hand returning upgraded brand reputation and revenue expansion.

## Key Deliverables

This project seeks to deliver the goals listed above. We can do so by integrating data from CDC studies, validating the predictions for accuracy, ongoing reassessments of app analytics and using real time data from frequent school surveys. The consistent up-keep with fresh and cleansed data results in personal interventions, proactive risk identification, product development, wellness initiatives and furthering partnerships. With these capabilities, It is natural that brand reputation enhances and provides expanded business application spans that captivate additional revenue streams.

The organizational decision-making this project aims to strengthen is fueled by streamlined data analysis and the actionable insights we can provide. By consolidating data from various sources and compiling them into a unified platform, data management and integration is possible along with its accessibility and quality assurance. Predictive, prescriptive and descriptive are the advanced analytics implemented to recommend actions and forecast trends. Real-time data visualizations will be available on our interactive dashboard used for exploration. This type of data empowers non-tech users by providing them the opportunity to generate insights and reports independent from IT.

These capabilities are what drives improved decision-making across departments because they are data-driven and agreed-upon faster—since everyone on the team has direct access to the data. Reducing manual reporting allows for an improved workflow, also depicted as operational efficiency that streamlines processes. Agility is a promise as the organization adapts to market changes with the real-time insights on the table.

## Project Timeline

Our project timeline should take approximately six months. We could start on January 5 and end on July 7, 2025. To expand on the key milestones, we can start with phase one that would last about one to two weeks and be completed by January 18. In this phase, we will aim to select one to two decisions to pilot and map these decisions by finding their frequency and how they impact businesses. We would also need to identify at least five top business-critical decisions.

Phase two would take three to six weeks—starting on January 21 with a completion date of February 15, 2025. During this time, identifying essential data sources will be our focus. Once established, basic dashboards can be created that offer actionable metrics. In doing so, establishing refreshing data schedules will also be executed. From there, we can test with actual decision-makers.

For our phase three, the duration will take two to six months in which we shall finish by July 7, 2025. Pilot decisions can be measured by their impact and documents will be created indicating successes and failures. Dependent the outcomes, the winning results can be applied to our next priority decisions. The steps will build our comprehensive BI infrastructure.

The testing, scaling and iterative development is safely packaged within this timeline. However,

stakeholder availability, team, size and complexity are aspects that would review to adjust our timeline appropriately.

## Proposed Approach

It is imperial to include a robust business intelligence architectures to achieve our project goals effectively. Our architectures are to be presented through intuitive analytics and reporting tools. To build them, we will be integrating data from various sources that can be transformed into our actionable insights.

Reporting and analytics, data sources, data storage, data management and data integration are a handful of the exponential components for the building blocks of our data architecture. Data sources connect and identify each other in flat files, transactional systems, and relevant databases. Tools like Talend, Azure Data Factory or Apache Airflow utilize data integration to extract, transform, and load (ETL) data into a centralized storage system. Data lakes (e.g., Azure Data Lake, Amazon S3), data warehouses (e.g., BigQuery, Snowflake) or lakehouses (e.g., Databricks) are popular data storage systems to effectively manage data. There are also a few popular BI tools that implement reporting and analytics like Looker for data visualization and analysis, Tableau or Power BI.

To ensure flexibility and scalability, harnessing a cloud-native architecture for our product and technology sector will be applied. Cost effective cloud-based services that are often used point to Google Cloud, AWS or Azure. For smooth maintenance, scalability and innovation enhancement, designing modular components is at hand. Trusted insights, established through data governance policies ensure data security and quality.

## Alternative Approaches

Although cloud-based BI platforms are preferred, a considerable alternative would be an on-premise solution. For comprehensive, proactive decision-making, predictive analytics driven by AI are an avenue to explore—especially combined with (DI) decision intelligence.

## Project Scope

The important business initiatives revolve around productivity and digital wellness. HR analytics, wellness program, development, and edtech partnerships make our project possible. In-scope deliverables for business are at high-level stakes and utilize predictive risk models, recommended interventions and screen time analytic dashboards. Aggregated data from wellness apps, existing HR systems, and edtech platforms, offer us data feeds. Data-driven decision-making is introduced through this project's BI tools and advanced analytics.

## Project Out-of-Scope

To govern a budget-friendly delivery that is on time, we set to address our clear boundaries. To mitigate invasive personal data, we decline to develop a consumer-facing app, individual screen time will not be tracked in real-time and our integration with device operating systems this will remain indirect. To focus resources on insight delivery and core analytics, we agreed upon constraints. With these guidelines, we believe our project objectives will be met within the time frame and budget set.

## PROJECT FINANCIALS

To analyze screen time effects on the employment outcomes of the future, the following summary provides the estimated breakdown of project costs. Please view the return on investment prediction along with other expected benefits below.

### Project Budget

#### Labor Costs (Expense)

- Internal Labor
  - o Data Analyst – \$12,000
  - o BI Developer – \$14,000
  - o Project Manager – \$8,000
  - o Subject Matter Expert – \$8,000

Subtotal Internal Labor: \$42,000

#### Technology & Tools (Capital/Expense depending on enterprise policy)

- Tableau licenses, cloud storage, analytics infrastructure – \$6,000

#### Data Acquisition (Expense)

- Data-sharing agreements, anonymization support – \$5,000

#### Contingency (Expense)

- Reserved for unexpected technical or data-related needs – \$4,000

Total Estimated Project Budget: \$57,000

## Total Cost of Ownership (TCO)

Beyond the initial project budget, the enterprise should anticipate recurring costs associated with maintaining the project's deliverables. These may include:

- Annual Tableau or cloud-platform subscription renewals
- Ongoing data-storage fees
- Periodic updates to dashboards, models, or data pipelines
- Support and maintenance from IT or analytics teams

These recurring costs contribute to the TCO and must be approved alongside the one-time project budget. Depending on enterprise policy, TCO may also be incorporated into ROI calculations.

## ROI

The estimated total project cost is \$57,000, including personnel, technology, data acquisition, and contingency. The project's benefits include improved decision-making for youth development programs, early identification of behavioral risk patterns, and long-term cost savings through targeted interventions. ROI is expected to be positive due to the high strategic value of understanding screen-time impacts on employability and the potential to reduce future workforce skill gaps.

ROI for this project is driven by:

- Improved program effectiveness: Effective early intervention is possible through data-driven insights by identifying at-risk populations for youth organizations and educators to focus on.
- Reduced long-term societal costs: When harmful screen-time patterns are identified early, future unemployment, skill deficits, and mental-health-related impacts are predicted to significantly reduce.
- Operational efficiency: Manual reporting time is reduced when analytics and dashboards are consolidated.
- Strategic value: A leader in future workforce readiness—by understanding digital-behavior trends, is the position I.Q. Productionz maintains when upholding this project.

While exact ROI calculations would be completed by the finance team, the combination of measurable operational benefits and high strategic importance strongly supports moving forward with the project.

## DEPENDENCIES, ASSUMPTIONS, RISKS AND ISSUES

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Every project faces uncertainties that could potentially derail success. While some might view this as simply covering all bases, identifying these factors early establishes critical warning systems that allow teams to respond proactively rather than reactively. When we anticipate challenges in advance, we create opportunities to mitigate their impact before they become problems.

### Project Dependencies

This project relies on several key dependencies both within our organization and from external partners. We need access to reliable datasets from schools, youth programs, and digital wellness organizations that have tracked screen-time patterns over meaningful periods. Our IT department must provide secure data storage and processing infrastructure to handle sensitive information appropriately. Research partners need to deliver behavioral and academic data according to agreed-upon timelines, and we'll depend on visualization tools like Tableau along with our analytical software platforms to process and present findings effectively. If any of these dependencies experience delays or limitations, our ability to deliver comprehensive analysis on time will be directly affected.

### Project Assumptions

Several key assumptions are aspects of our project plan. One of our biggest assumptions relies heavily on screen-time data received being complete, accurate and measured consistently across different sources. By expecting stakeholders to remain active with continued engagement throughout the project lifecycle defines it as an assumption--even with effective follow-ups. To decrease this assumption, we can draft an agreement to be signed to assure follow-through. This includes holding stakeholders accountable to provide guidance and feedback when or where it is needed. Our technical infrastructure aims to enable effective visualization of complex patterns and support the integration of multiple data streams. Additional dependencies encompass sufficient in-house expertise to analyze the intersections between academic performance, behavioral patterns and employment-related activities. Lastly, our timeline has been coordinated with the expectation that our external partners will deliver their data within mutually agreed timeframes. If any of these assumptions prove incorrect, we'll need to reassess our approach and potentially adjust our timeline, resource allocation or scope accordingly.

## Project Risks

*We've identified several risks that warrant careful monitoring throughout the project:*

*Data Quality Risk* remains our highest concern. To enhance the practical value of our findings, our conclusions are only secured with consistent, complete data. With meticulous watch for warning signs that warrant inconsistencies, we're implementing rigorous data-cleaning protocols and validation checks at each stage of processing and ingestion. Our red flags wave at measures and attributes that have missing fields, irregular timestamps, or conflicting metrics across different sources.

*Stakeholder Misalignment* could emerge if sponsors shift priorities mid-project. Reduced meeting attendance, delayed approvals, or changing feedback patterns can alert us to take action immediately. By using early indicators like those above, we can initiate regular communication that involves transparent milestone reviews to realign our goals. It is inevitable that stakeholders need extra explanations to understand both progress and challenges as they develop.

*Interpretation Risk* presents a particular challenge given the complexity of our research question. Commonly, misused findings can support claims about causal relationships between screen time and employment outcomes. This effect can be decreased by appropriate statistical disclaimers, providing clear documentation and sufficient context for readers overstand what the data shows and what it doesn't.

*Technical Risk* materialization is alleviated when our tools are robust and integration is on point. Technical stack proactive pretesting pins-down on-time delivery of analysis by identifying backup tools. It also keeps strong coordination with IT support staff.

## Project Issues

Several ongoing concerns beyond the risks outlined above, merit our attention. The most fundamental challenge is the limited availability of longitudinal data that directly connects screen-time patterns to employment outcomes—most existing datasets weren't designed with this specific question in mind. Additionally, variability in self-reported screen-time metrics may reduce precision, as different individuals interpret and report their usage quite differently. We also recognize that some stakeholders may interpret correlations as causation despite our best efforts to communicate the limitations of observational data. Finally, we anticipate potential resistance from groups who view screen-time concerns as either overstated or politically charged, which could complicate stakeholder engagement and the reception of our findings.

## BUSINESS SPONSORS AND STAKEHOLDERS

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Establishing the right sponsorship structure early-on in our project will be essential to navigating the challenges outlined above and ensuring our findings receive appropriate consideration and application. Careful selection of has been agreed upon below and we are open to any additional selections or changes as see fit prior to the project start date.

### Business Sponsors

**Dr. Melissa Carter** – Director of Workforce Development, FutureReady Institute

Project Role: Executive Sponsor

Rationale: Owns strategic initiatives related to youth employability and long-term workforce readiness. Provides funding approval, organizational alignment, and executive-level support.

**James Patel** – Chief Data & Insights Officer, Youth Digital Wellness Coalition

Project Role: Data Sponsor

Rationale: Oversees data governance, analytics strategy, and ensures that the project aligns with broader digital-wellness research initiatives.

### Project Stakeholders

**Sarah Nguyen** – Program Manager, Digital Literacy Programs

Role: Operational Stakeholder

Interest: Uses project findings to shape curriculum and interventions.

**Dr. Aaron Lopez** – Senior Research Analyst, Behavioral & Educational Outcomes Lab

Role: Analytical Stakeholder

Interest: Applies insights to ongoing research on screen-time impacts.

**Kendra Williams** – Community Outreach Coordinator, Future Ready Institute

Role: Engagement Stakeholder

Interest: Communicates findings to parents, schools, and community partners.

## APPROVALS

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| Title             | Name               | Approval<br>Date |
|-------------------|--------------------|------------------|
| Executive Sponsor | Dr. Melissa Carter |                  |
| Data Sponsor      | James Patel        |                  |
| BI Director       | Kathy Junebug      |                  |
| Director of HR    | Jennifer Martinez  |                  |
| Project Manager   | Sarah Nguyen       |                  |

## APPENDIX: BACKGROUND MATERIALS

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### *Project Plan & Detailed Timeline*

A comprehensive Work Breakdown Structure (WBS) has been developed for the six-month period (January 5 – July 7, 2025). The plan details the three-phase approach, including the "Decision Pilot" selection in Phase 1 and the full BI infrastructure rollout in Phase 3.

### *Project Budget (TCO Analysis)*

Total investment is estimated at **\$57,000**. Beyond initial labor and technology costs, the Total Cost of Ownership (TCO) includes recurring cloud storage fees, Tableau subscription renewals, and periodic model retraining to ensure the predictive analytics remain accurate as digital usage patterns shift.

### *ROI Calculation*

Financial modeling based on a 15% improvement in workforce uptime and a reduction in mental-health-related absenteeism. The ROI is supported by the strategic value of brand protection and the creation of new revenue-generating EdTech partnerships.

## Authors and contributors

- **Hali Sarah Parsons (Lead Author):** Primary BI Strategist and document architect. Responsible for the Business Justification and Technical Requirements alignment.
- **Dr. Melissa Carter (Executive Sponsor):** Provided strategic vision for youth employability and long-term workforce impact metrics.
- **James Patel (Data Sponsor):** Provided data governance standards and access to the Youth Digital Wellness Coalition datasets.
- **Jennifer Martinez (HR Director):** Contributed "at-risk" behavioral indicators and employee productivity KPI definitions.
- **Dr. Amanda Foster (SME):** Clinical Psychologist who provided the logic for the mental health risk scoring models.
- **Christopher Lee (SME):** Data Architect who designed the cloud-native integration paths for the CDC and wellness app feeds.